PROBLEM SET 4

Hamming Codes and Boolean algebra

~~1. Correct the following bit sequence for even parity bits using Hamming code~~
~~1011 1010 1001 0100 1010 0101 0001 1110 0011 1010~~ **~~(optional to be done)~~**

2. Correct the following for odd parity bit sequence using Hamming code:
(a) 10110101001
(b) 11011101

3. Verify by the truth table method.
(a) $A+\overline{A}B + AB = A+B$
(b) $(A + \overline{B})(\overline{A} + B) = AB + \overline{A}\,\overline{B}$

4. Reduce the following Boolean expressions.
(a) $XY + XY + \overline{XY}$
(b) $XYZ + YYZ + \overline{Y}Z + \overline{X}Y$

5. Reduce the following Boolean expressions.
(a) $\overline{B}\,\overline{C}\,D + (\overline{B + C + D}) + \overline{B}\,\overline{C}\,\overline{D}E$
(b) $\overline{\left(A\,\overline{B}\,C\right)\left(\overline{A\,\overline{B}}\right) + BC}$
(c) $(WX + W\overline{Y})(X + W) + WX(\overline{X} + \overline{Y})$

6. Prove that
(a) $ABCD + AB(\overline{CD}) + \overline{(AB)}CD = AB + CD$
(b) $(A + \overline{A})(AB + AB\overline{C}) = AB$
(c) $(A\overline{B} + A\overline{C})(BC + B\overline{C})(ABC) = 0$
(d) $A\overline{B}C + \overline{A}BC + ABC = C(A + B)$

7. Apply De Morgan's theorem to each of the following expressions.

(a) $\overline{P\,(Q + R)}$ (b) $\overline{\left(P + \overline{Q}\right)(\overline{R} + S)}$

(c) $\overline{\overline{(A + B)}\;\overline{C + D)}}\;\overline{\overline{(E + F)}\;\overline{(G + H)}}$ (d) $\overline{\left(\overline{A} + B + \overline{C} + D\right)(0 + \overline{A}BC\overline{D})}$

8. Without reducing, convert the following expressions to NAND logic.

(a) $AB + CD\,(A\overline{B} + CD)$

9. Without reducing, convert the following expressions to NAND logic.
(a) $(XY + Z)(XY + P)$ (b) $(1 + A)(ABC)$

10. Without reducing, convert the following expressions to NOR logic

(a) $X\overline{Y} + X + \overline{X+Y}$ (b) (1 + A) (AC)

11. Without reducing, implement the following expressions in AOI logic and then convert them into (1) NAND logic and (2) NOR logic.

(a) $(A + \overline{B}C) + D$ (b) $A + B\,\overline{C} + \overline{B+C} + \overline{B}\,\overline{C}$